

Hellenic Complex Systems Laboratory

Analysis of Diagnostic Accuracy Measures

Technical Report IX

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2015

Analysis of Diagnostic Accuracy Measures

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Short Description of the Demonstration

This Demonstration shows plots of various accuracy measures for diagnostic tests in normally distributed nondiseased and diseased populations. This is done for different values of disease prevalence, taking into account the means and standard deviations of the populations. The means and standard deviations are expressed in arbitrary units. The measures considered are the positive predictive value (PPV), the negative predictive value (NPV), the (diagnostic) odds ratio (OR), the likelihood ratio for a positive test result (LR+), and the likelihood ratio for a negative test result (LR-). The measures are plotted against the sensitivity or the specificity of the test, which can be selected by clicking the respective button.

Search Terms

sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, normal distribution, binormal distribution, positive predictive value, negative predictive value, likelihood ratio, odds ratio

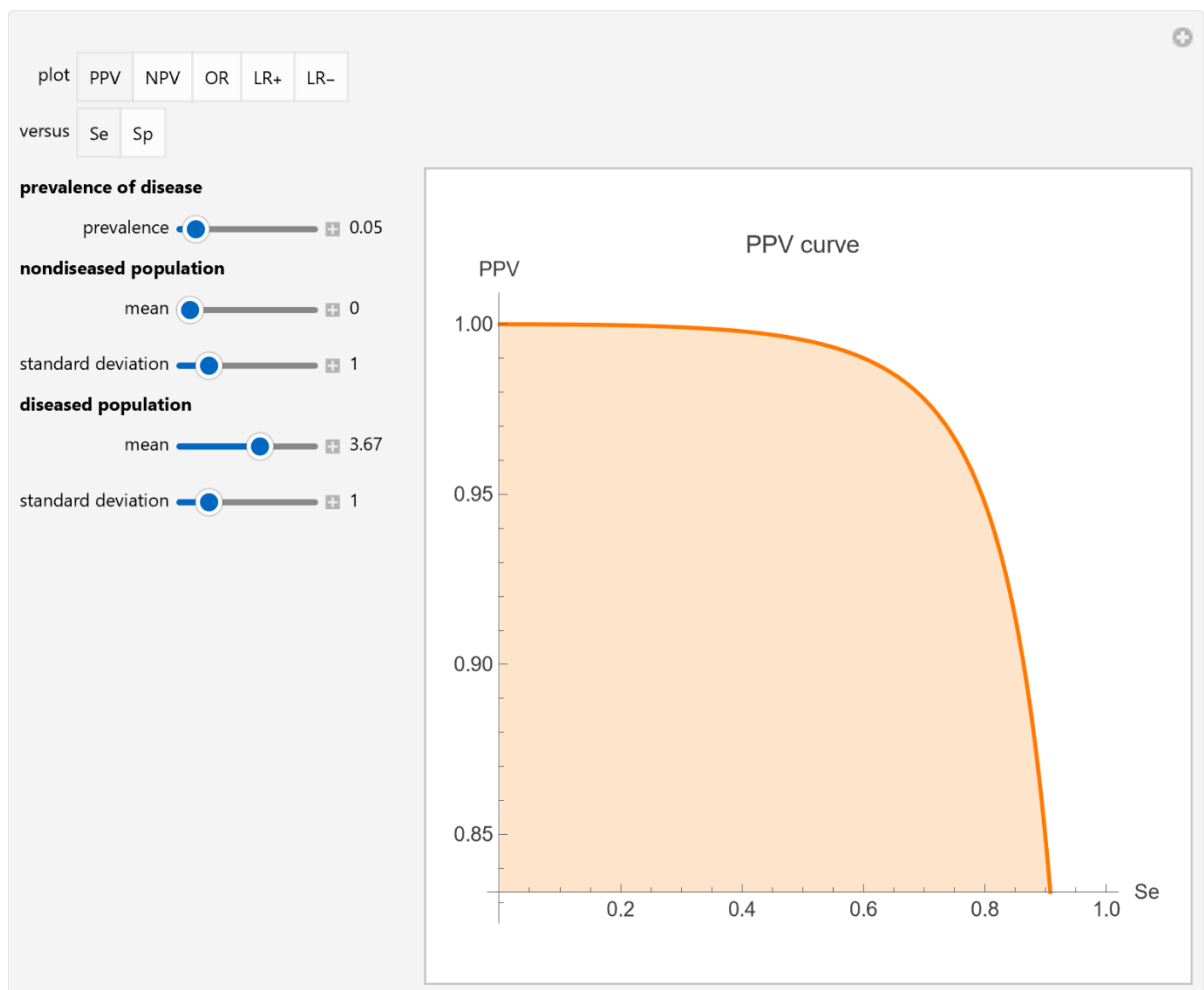


Figure 1: Positive predictive value vs specificity curve plot of a diagnostic test, with the settings shown on the left.

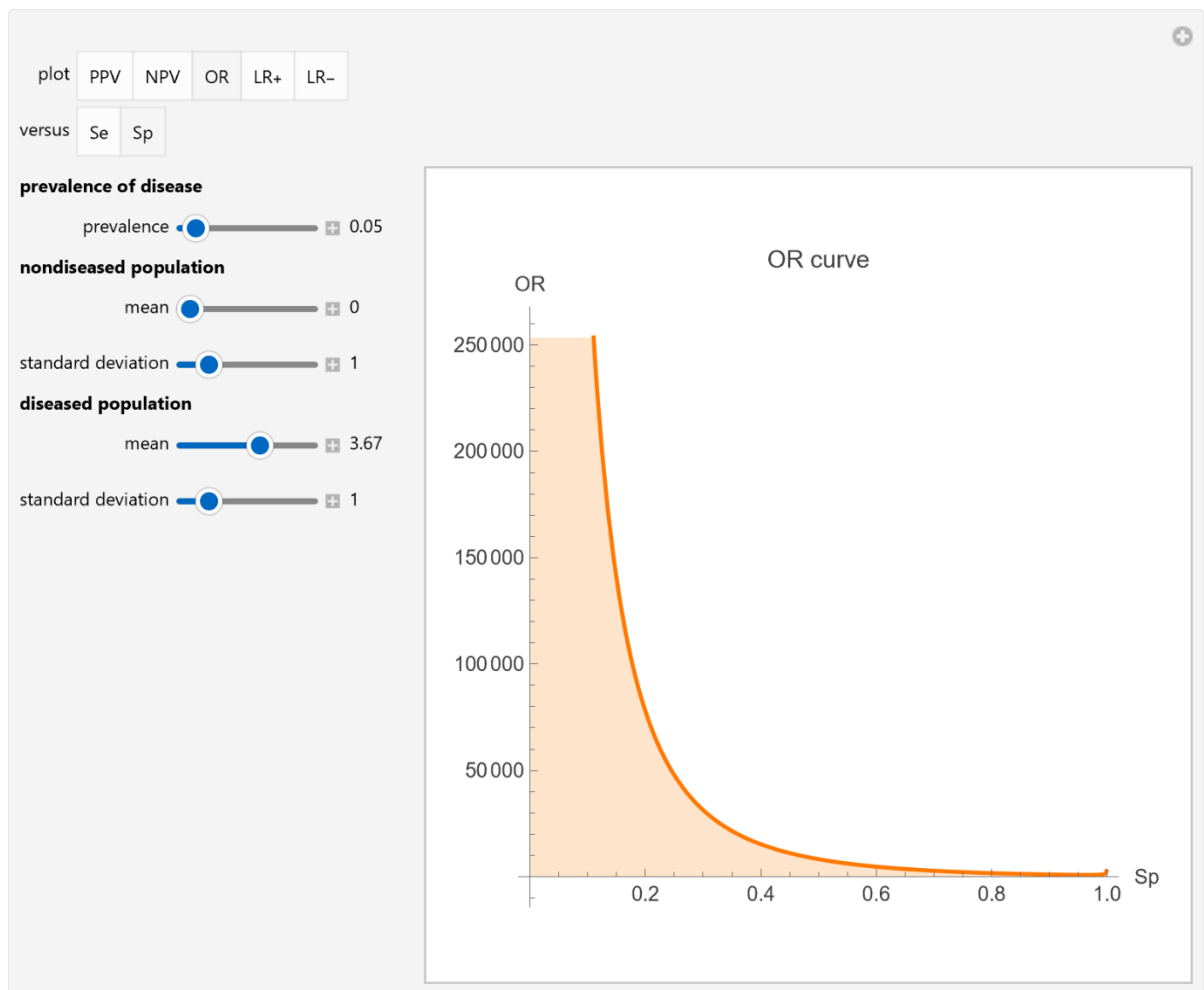


Figure 2: Odds ratio vs specificity curve plot of a diagnostic test, with the settings shown on the left.

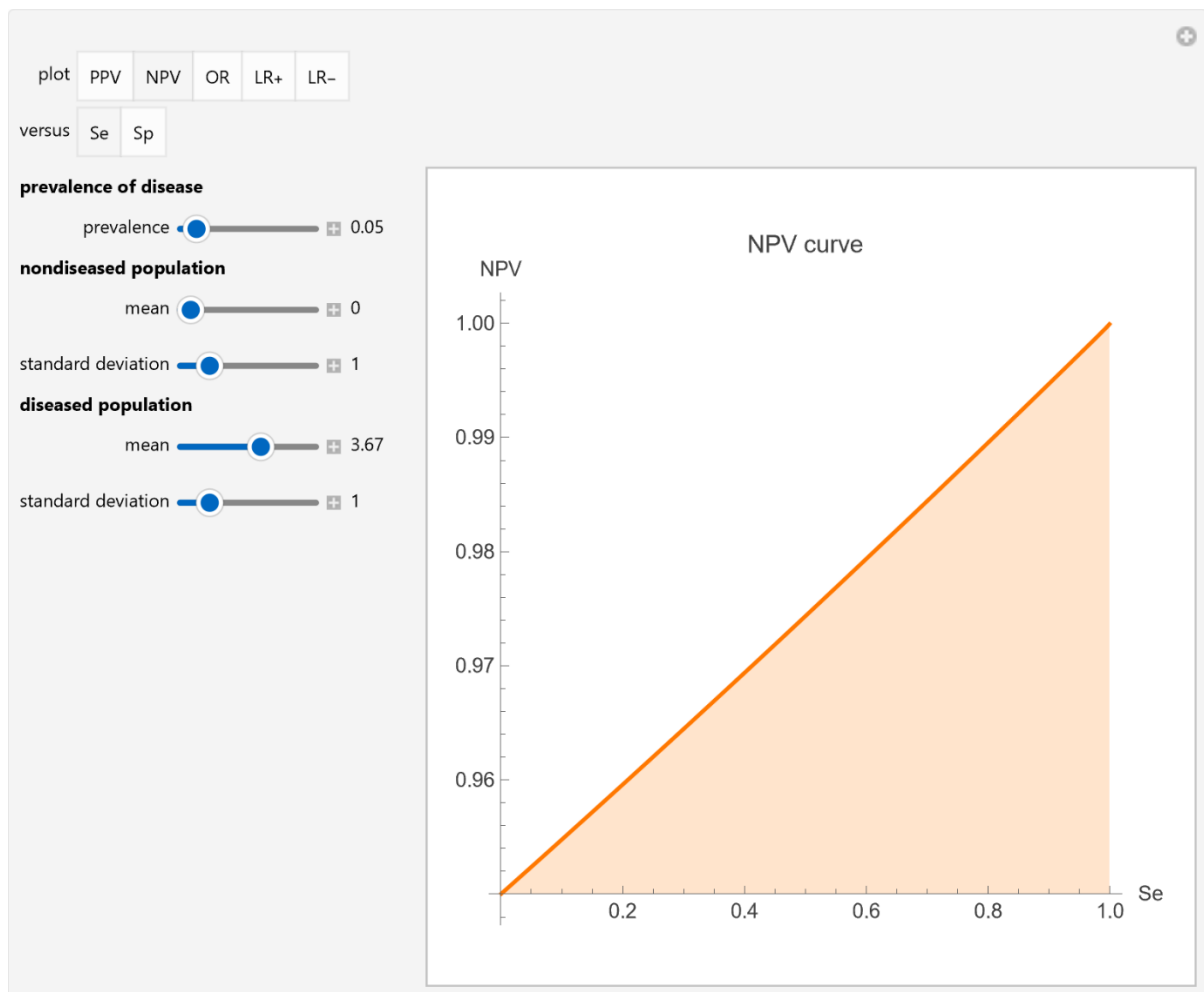


Figure 3: Negative predictive value vs sensitivity curve plot of a diagnostic test, with the settings shown on the left.

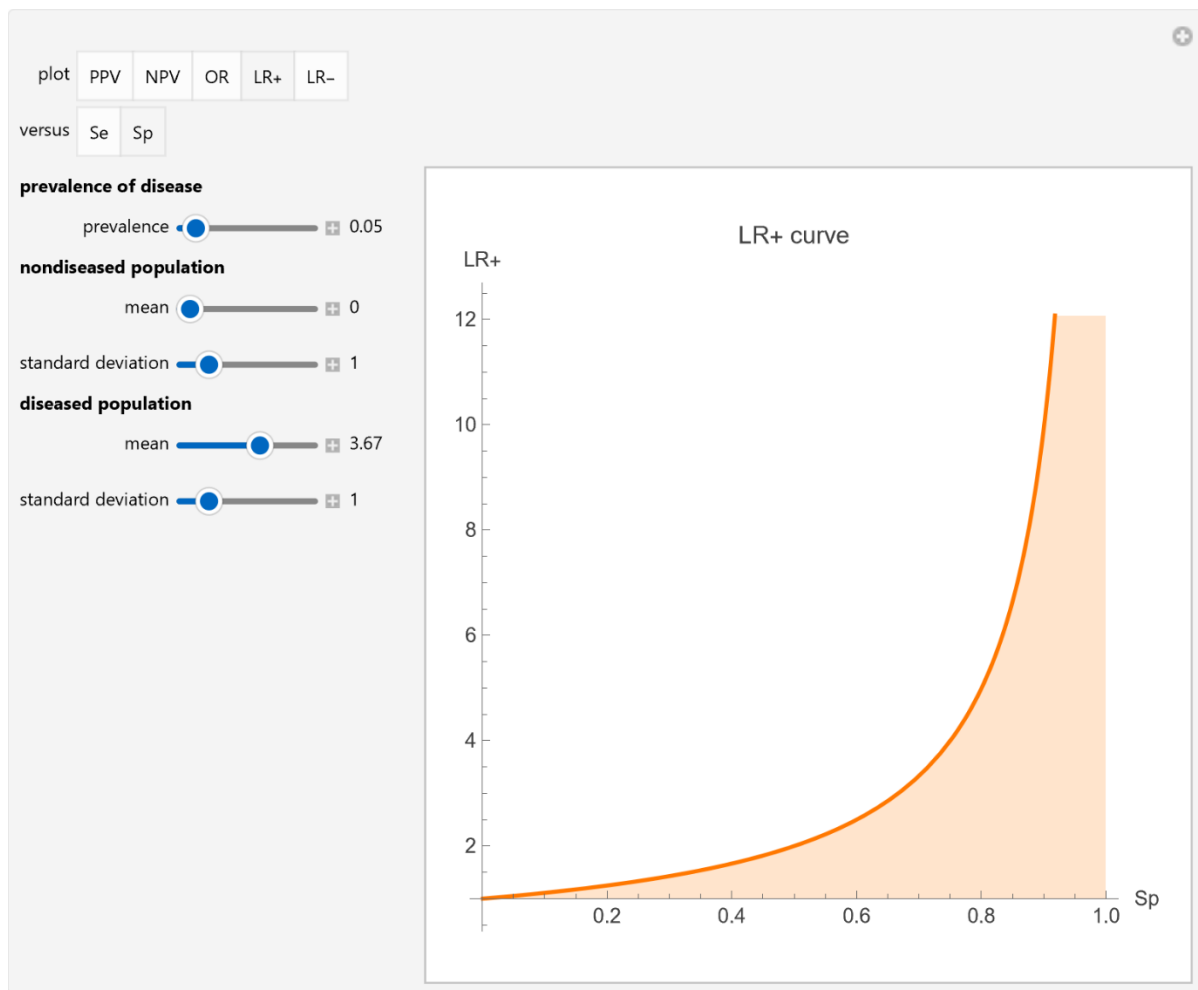


Figure 4: Likelihood ratio for a positive result vs specificity curve plot of a diagnostic test, with the settings shown on the left.

Details

The measures that are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased or nondiseased population can be calculated as functions of the sensitivity or the specificity of the test. Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. If we denote by Se the sensitivity, Sp the specificity, and v the prevalence, we have:

$$PPV = \frac{Se v}{Se v + (1 - Sp)(1 - v)}$$
$$NPV = \frac{Sp (1 - v)}{Sp (1 - v) + (1 - v) pr}$$
$$OR = \frac{\frac{Se}{1 - Se}}{\frac{1 - Sp}{Sp}}$$
$$LR+ = \frac{Se}{1 - Sp}$$
$$LR- = \frac{1 - Se}{Sp}$$

In the figures, the population data describe a bimodal distribution of serum glucose measurements in nondiabetic and diabetic populations [2].

This Demonstration is a simplified version of another Demonstration [1], and is appropriate as an educational tool appropriate as an educational tool in medical statistics and diagnostic test evaluation.

References

- [1] A. T. Hatjimihail. Uncertainty of Measurement and Diagnostic Accuracy Measures. Ver. 1.1.1. Hellenic Complex Systems Laboratory; 2026. Available at: <https://www.hcsl.com/Tools/Demonstrations/UncertaintyOfMeasurementAndDiagnosticAccuracyMeasures.nb>
- [2] T. O. Lim, R. Bakri, Z. Morad, and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care*, **25**(12), 2002 pp. 2212–2217.

Demonstration

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Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/AnalysisOfDiagnosticAccuracyMeasures.nb>

First published in 2015 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/):

<https://demonstrations.wolfram.com/AnalysisOfDiagnosticAccuracyMeasures/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

Chatzimichail T. *Analysis of Diagnostic Accuracy Measures*. Ver. 1.1.1. Hellenic Complex Systems Laboratory; 2026.

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Citation

Chatzimichail T. *Analysis of Diagnostic Accuracy Measures*. Hellenic Complex Systems Laboratory; 2007.

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First Published: July 24, 2015

Revised: May 23, 2026